Data Report on Earthquakes

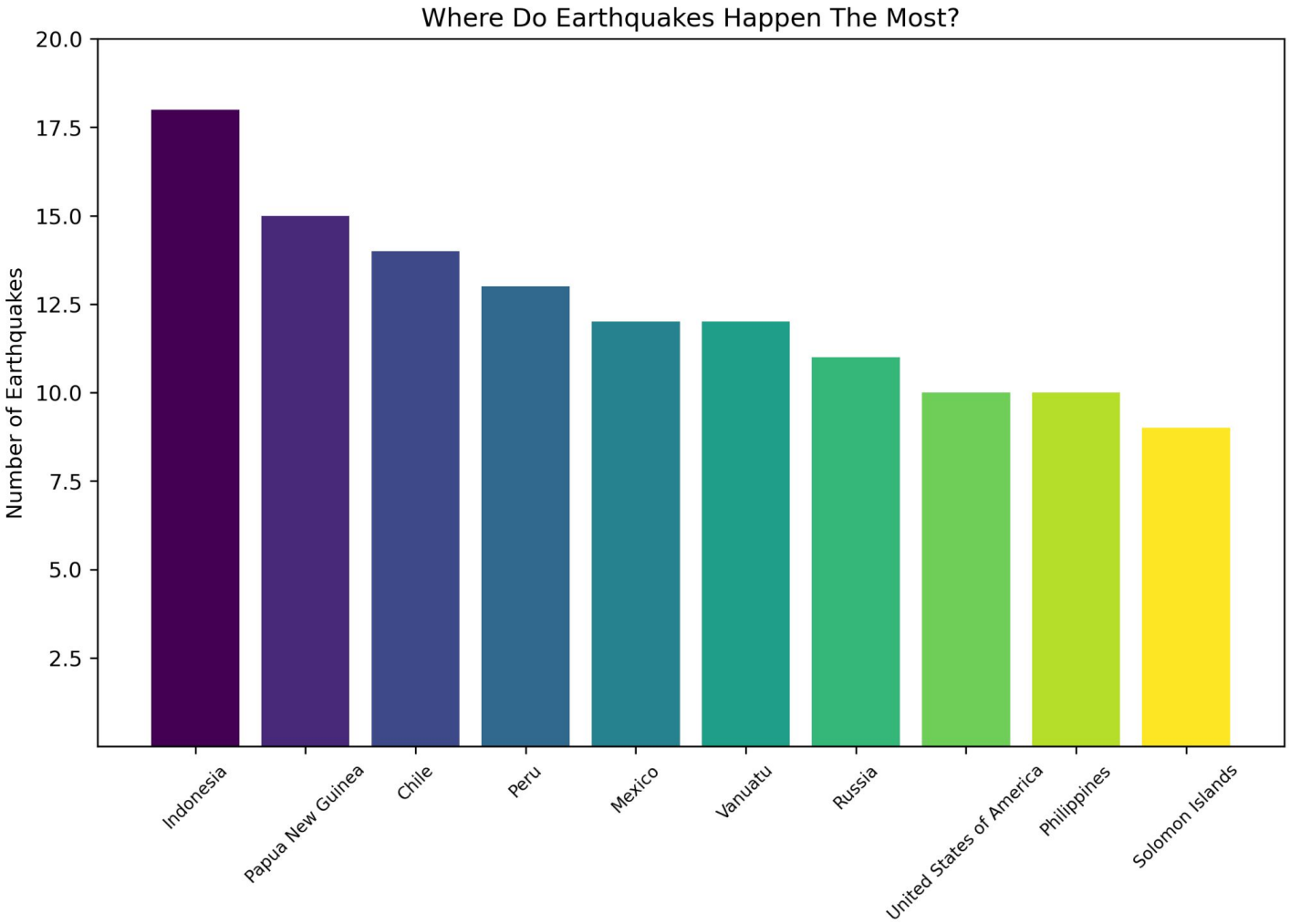
The Data

The Earthquake data spans from years 1995 - 2023.

- This dataset contains over 1000 documented Earthquakes.
- All continents are included.
- Everywhere from magnitude, magnitude type, and alert levels are included in the data recordings.

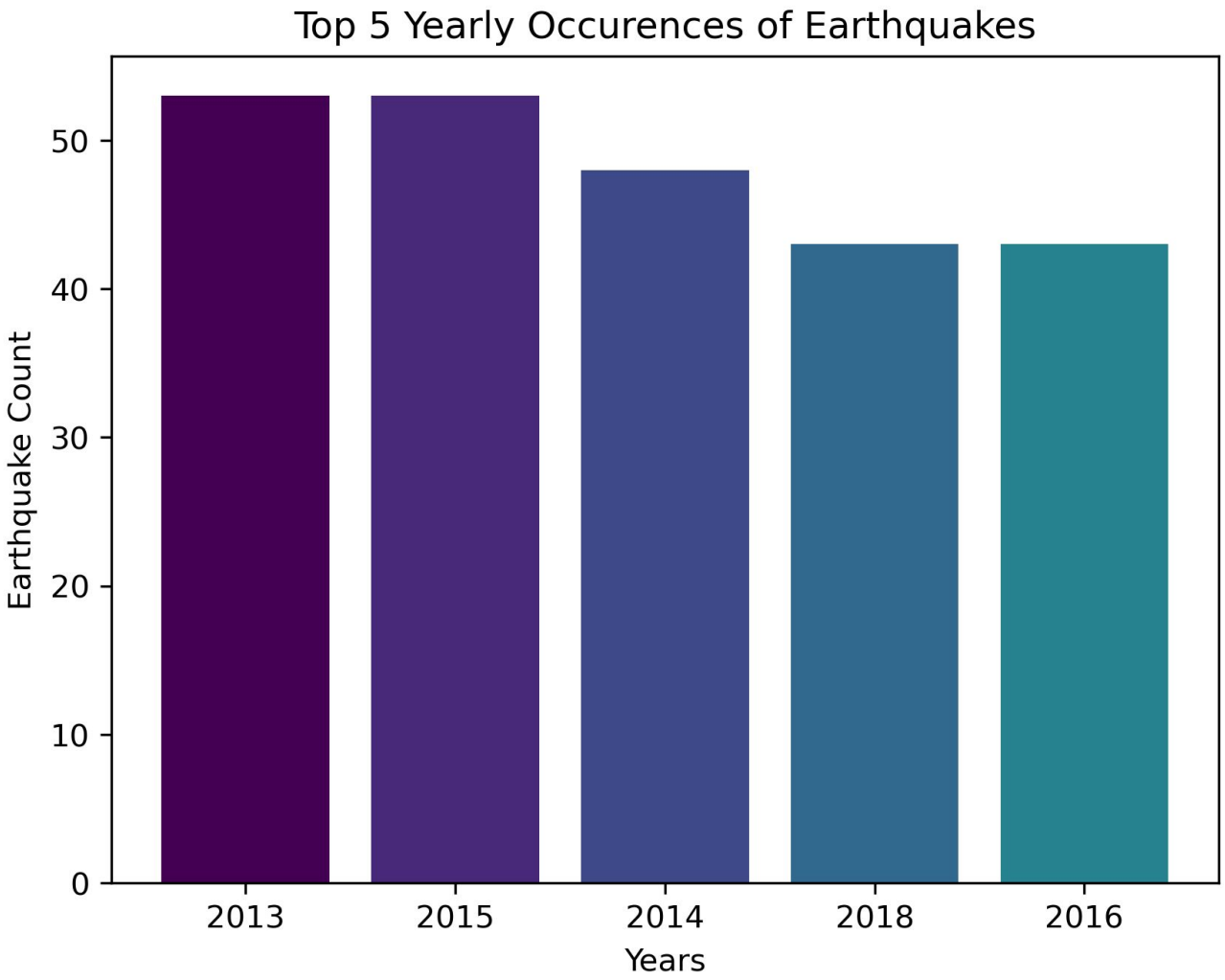
Where Do Earthquakes Happen The Most?

Firstly, knowing where earthquakes happen the most provides a basic understanding of our data. To see the actual graph, please skip to the next slide.


Where Do Earthquakes Happen The Most?
Number of Earthquakes
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
Indonesia
Papua New Guinea
Chile
Peru
Mexico
Vanuatu
Russia
United States of America
Philippines
Solomon Islands

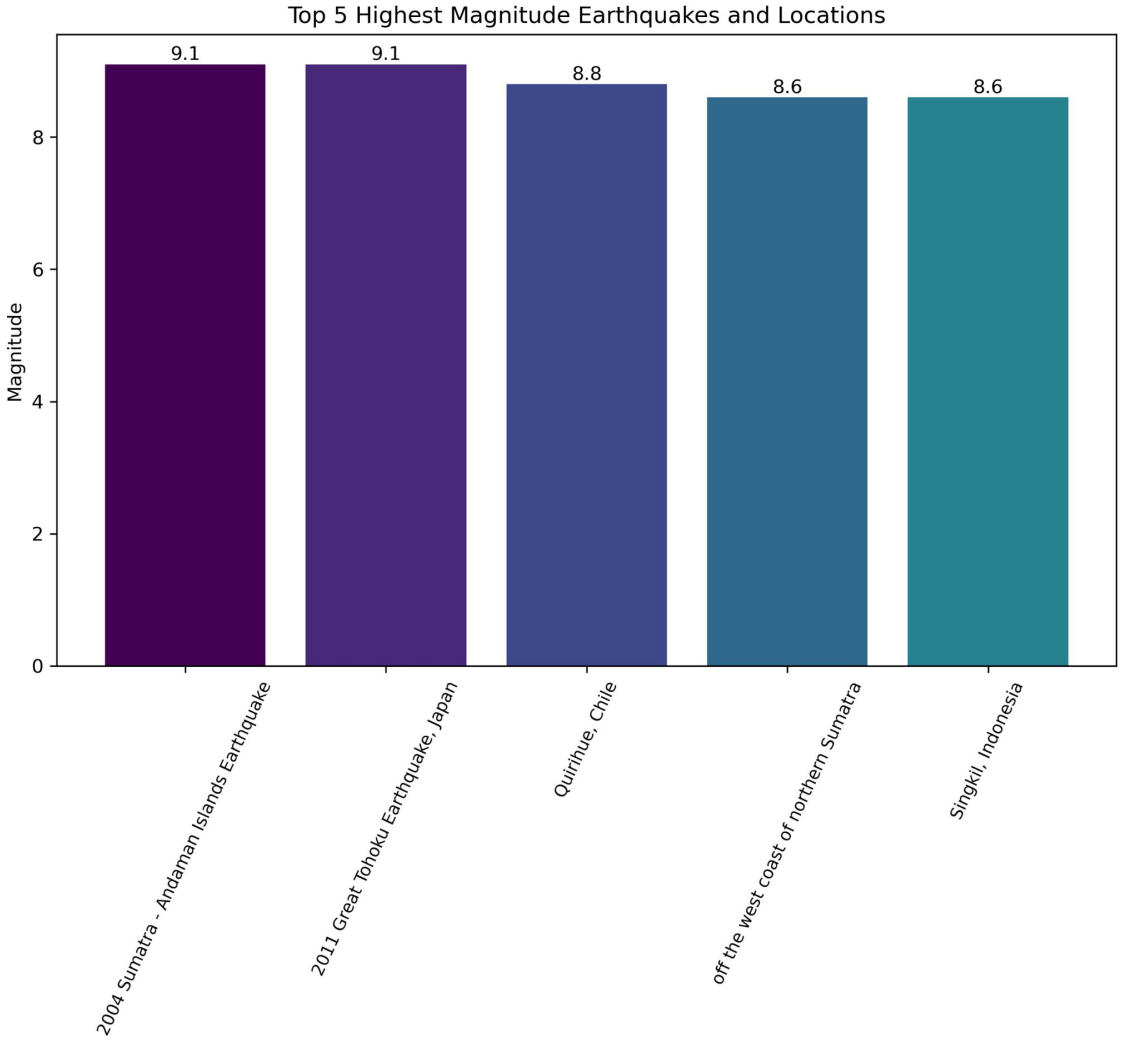
What Years Do Earthquakes Happen The Most?

Now that we know where all of the Earthquakes tend to happen, the next question is when do they happen? Kindly skip to the next slide to see the data.


Top 5 Yearly Occurences of Earthquakes
Earthquake Count
0
10
20
30
40
50
2013
2015
2014
2018
2016
Years

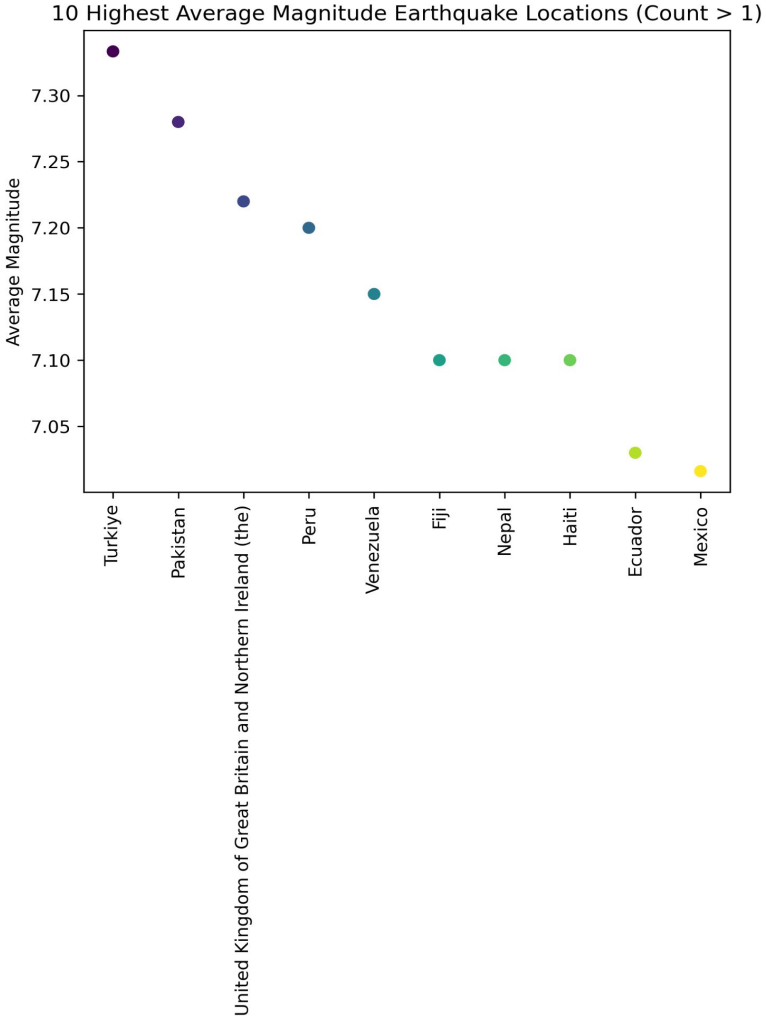
What Is Considered Most Dangerous?

So we figured out where Earthquakes mostly occur and the years when Earthquakes happen the most. The next question to ask is, where are the highest magnitude Earthquakes?


Top 5 Highest Magnitude Earthquakes and Locations
Magnitude
0
2
4
6
8
9.1
9.1
8.8
8.6
8.6
2004 Sumatra - Andaman Islands Earthquake
2011 Great Tohoku Earthquake, Japan
Quirihue, Chile
off the west coast of northern Sumatra
Singkil, Indonesia

Last Question

The final question that was drawn is, where do the *highest* magnitude earthquakes happen on average? Knowing what we know about the highest magnitude earthquakes, do locations correlate or do earthquakes with high magnitudes happen at random?


10 Highest Average Magnitude Earthquake Locations (Count > 1)
Average Magnitude
7.30
7.25
7.20
7.15
7.10
7.05
Turkiye
Pakistan
United Kingdom of Great Britain and Northern Ireland (the)
Peru
Venezuela
Fiji
Nepal
Haiti
Ecuador
Mexico

Conclusion

The data shows that overall earthquakes do not happen in a predictable manner. From where the most earthquakes happen to the highest magnitudes, the most prominent data metrics are always changing.

So, earthquakes truly are a freak of nature. Though with more questions and data we can hope to learn more about their nature.

Thank You For Your Time!